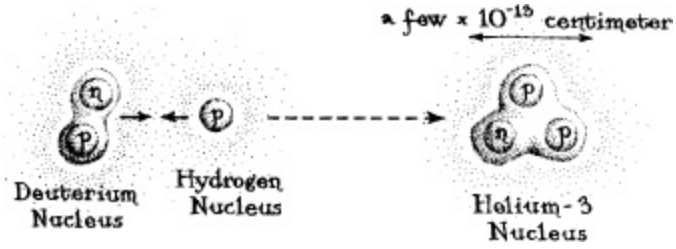
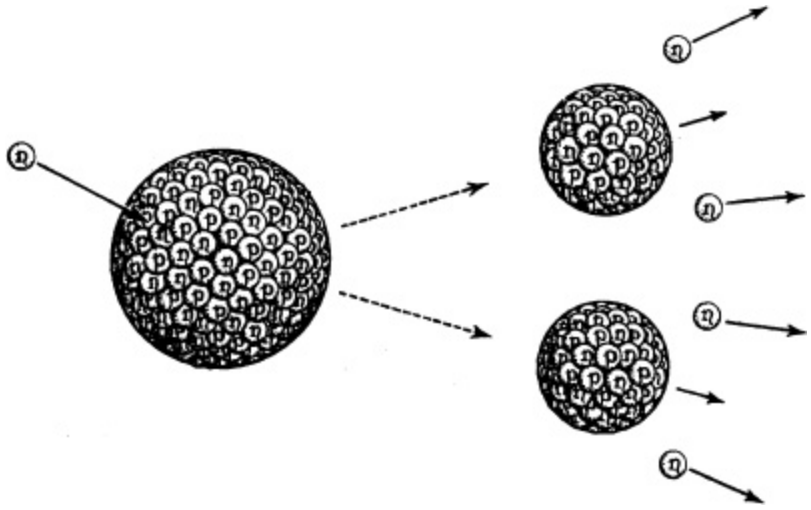
nucleus (two protons and one neutron): Such fusion reactions keep the Sun hot and power the hydrogen bomb (the “superbomb” as it was called in the 1940s and 1950s).



The *fission* (splitting apart) of a very heavy nucleus to form two medium-sized nuclei releases a large amount of energy—far more than comes from chemical reactions (since the nuclear force which governs nuclei is far stronger than the electromagnetic force which governs chemically reacting atoms), but much less energy than comes from the fusion of light nuclei. A few very heavy nuclei undergo fission naturally, without any outside help. More interesting for this chapter are fission reactions in which a neutron hits a very heavy nucleus such as uranium-235 (a uranium nucleus with 235 protons and neutrons) and splits it roughly in half:



There are two special, heavy nuclei, uranium-235 and plutonium-239, with the property that their fission produces not only two medium-sized nuclei, but also a handful of neutrons (as in the drawing above). These neutrons make possible a *chain reaction:* If one concentrates enough uranium-235 or plutonium-239 into a small enough package, then the